

1. In the following quiz, you will practice using the chain rule, which allows us to differentiate functions of functions. We saw in the previous videos that for two functions $g(h)$ and $h(x)$, the derivative of g with respect to x is given by $\frac{dg}{dx} = \frac{dg}{dh} \frac{dh}{dx}$, where the two derivatives on the right hand side are in some way 'chained together'.

1 point

If $f(x) = g(h(x))$, which of the following is the equivalent to the chain rule in the $f'(x)$ notation?

- ☐ $f'(x) = g'(h'(x))$
- ☐ $f'(x) = g'(h'(x))h'(x)$
- ☐ $f'(x) = g'(h(x))$
- ☒ $f'(x) = g'(h(x))h'(x)$

2. Much like the product rule, the art of the chain rule lies in identifying the components of the function that allow you to apply the rule.

1 point

Consider the function $f(x) = e^{x^2-3}$. We can break up $f(x)$ by writing $g(h) = e^h$ and $h(x) = x^2 - 3$. Now $f(x) = g(h(x))$

Use the chain rule to calculate $f'(x) = \frac{df}{dx}$.

- ☐ $f'(x) = e^{x^2-3}$
- ☐ $f'(x) = 2e^{x^2-3}$
- ☒ $f'(x) = 2xe^{x^2-3}$
- ☐ $f'(x) = (x^2 - 3)e^{x^2-3}$

3. Use this same process to identify the functions that make up $f(x) = \sin^3(x)$, and calculate $f'(x)$.

1 point

- ☐ $f'(x) = \cos^3(x)$
- ☒ $f'(x) = 3\sin^2(x)\cos(x)$
- ☐ $f'(x) = 3\sin^2(x)$
- ☐ $f'(x) = 3\cos^2(x)$

4. Now you will tame a beast of your own by calculating the derivative of $\tan(x)$ with respect to x . You will need to use both the product rule and the chain rule.

1 point

If you want to try it on your own you can ignore the following hints, but they might be useful if you'd like some help starting.

Hint 1: The first step is to use the trigonometric relation $\tan(x) = \frac{\sin(x)}{\cos(x)}$.

Hint 2: The next step is to remember that $\frac{1}{\cos(x)} = [\cos(x)]^{-1}$. Now try to identify the functions that will allow you to calculate $\frac{d}{dx} \left(\frac{1}{\cos(x)} \right)$.

- ☐ $\frac{d}{dx} \tan(x) = 1 - \frac{\sin(x)}{\cos^2(x)}$
- ☐ $\frac{d}{dx} \tan(x) = 1 - \tan^2(x)$
- ☒ $\frac{d}{dx} \tan(x) = 1 + \tan^2(x)$
- ☐ $\frac{d}{dx} \tan(x) = \tan^2(x)$
- ☐ $\frac{d}{dx} \tan(x) = 0$

5. The chain rule can also be applied to functions of functions of functions. Consider a function $f(g(h(x)))$. By applying the chain rule twice, it's possible to show that $\frac{df}{dx} = \frac{df}{dg} \frac{dg}{dh} \frac{dh}{dx}$. Notice how the derivatives chain together! If you'd like you can try to show this result yourself.

1 point

Use this information to find the derivative of $f(x) = e^{\sin(x^2)}$ with respect to x . Remember to write down the appropriate functions $f(g)$, $g(h)$ and $h(x)$ to get started.

- ☒ $f'(x) = 2xe^{\sin(x^2)} \cos(x^2)$
- ☐ $f'(x) = 2e^{\sin(x^2)} \sin(x) \cos(x)$
- ☐ $f'(x) = e^{\sin(x^2)} \cos(x^2)$
- ☐ $f'(x) = 2xe^{\sin(x^2)}$